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Scientific Article

Descriptive and temporal analysis of post-mortem lesions recorded in slaughtered pigs in New Zealand from 2000 to 2010

EJ Neumann^{*§}, WF Hall[†], MA Stevenson^{*}, RS Morris[‡] and J Ling Min Than[#]

Abstract

AIMS: To complete a retrospective analysis of data from a national abattoir-based lesion recording system (PigCheck) in the New Zealand pig industry, in order to establish the prevalence of 20 post-mortem disease lesions, describe long-term trends in the prevalence of these lesions, and identify the proportion of the monthly variation in lesion prevalence that could be attributed to individual farms or abattoirs.

METHODS: Slaughter lesion data were collected and reported at the lot level (a cohort of pigs delivered from one farm, at one time). Data on the prevalence of lesions between January 2000 and December 2010 was aggregated by month, and time-series analysis of the data for each lesion was conducted. The time series pattern for each lesion was described with an auto-regressive integrated moving average (ARIMA) model; seasonality of lesion occurrence was assessed separately. To determine the proportion of variance in lesion prevalence that could be attributed to farms relative to that attributed to abattoirs, a hierarchical binomial generalised linear mixed model was created incorporating two random effect levels, at the farm (within abattoir) and abattoir levels.

RESULTS: A dataset comprised of 124,407 lots (6,220,664 pigs, 279 farms, five abattoirs) was compiled for analysis. The most prevalent conditions across the 11-year time series were antero-ventral pneumonia (7.6%), pleuropneumonia (11.4%), and milk spots (9.2%). Of the 15 lesions shown to have a significant annual change in prevalence, 10 decreased over time and five increased. The variance in prevalence that was observed for pyogenic lesion (92%), mange (73%), and ileitis (62%) was attributed primarily to variation between abattoirs. By contrast, the farm of origin explained the greatest percentage of variance in prevalence for rectal prolapse (98%), pneumonia (97%), and antero-ventral pneumonia (96%).

CONCLUSIONS: The overall prevalence of most lesions recorded in PigCheck for the period was low relative to published data from other countries. Common lung pathologies contributing to lesions such as antero-ventral pneumonia and pleuropneumonia were primarily a function of farm management and were not likely due to variability in lesion recording at different abattoirs.

CLINICAL RELEVANCE: Based on the low frequency of lesions in pigs at commercial abattoirs, the health status of pigs in the New Zealand pig industry is considered to be very good. Pneumonia, pleurisy, and ascariasis are some of the most prevalent conditions that should be focussed on through development of herd health management plans.

KEY WORDS: *Pig; epidemiology; population surveillance; abattoirs; pneumonia*

Introduction

Programmes have been established in many countries around the world to collect data at abattoirs that describe the presence of lesions or disease agents in pig carcasses that are associated either with poor meat quality or that present a risk to human health. Less frequent are standardised and on-going programmes to collect information from abattoirs that can inform producers about the presence or frequency of diseases that are important because of their negative impact on cost of production, rather than their relationship to food-borne diseases of people. Such national programmes are known to exist in some countries including Great Britain (Sanchez-Vazquez *et al.* 2011), the Netherlands (Blocks *et al.* 1994), Scandinavian countries (Olsson *et al.* 2001) and New Zealand (Anonymous 1999), with additional informal, producer-driven efforts likely occurring in most other countries with modern pig industries. However these informal “slaughter-checks” tend to be periodic rather than on-going, are targeted at specific known problems rather than comprehensively surveying a broad range of disease conditions, are customised to a specific farm’s or veterinarian’s needs rather than approached systematically, and their results are not routinely published.

The New Zealand Pork Industry Board (NZ Pork), a statutory board established by the Pork Industry Board Act (1997) to represent the interest of pig farmers, developed a formal national programme to survey the presence of important production limiting

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Table 1. Description of 20 lesions reported in a national abattoir-based lesion recording system (PigCheck) between 2000 and 2010 in New Zealand

Lesion name	Definition
Arthritis	Swelling and/or excess fluid associated with tarsal, metatarsal, carpal, or metacarpal joints; may include suppurative or non-suppurative inflammatory responses.
Broken back	Vertebral fracture caused by inappropriate electrical stunning conditions.
Bruising	Excoriation or bruising of the skin.
Emaciation ^a	Subjective assessment based prominent bony protuberances and ribs.
Full gut ^a	Stomach contains appreciable (>250 mL) quantity of feed or ingesta.
Ileitis	Grossly visibly thickening or haemorrhage of ileal wall.
Kidney condition	Presence of haemorrhage, white spots, or chronic swelling of kidney(s).
Mange	Red papules, predominantly on ears, belly, or rump/scrotum.
Milk spot	Presence of white spots on surface of liver as a result of roundworm larvae migration.
Other lesions	Miscellaneous occurrence of infectious or non-infectious disease processes not otherwise listed.
Pericarditis ^a	Excessive accumulation of fluid around the heart with or without pericardial thickening and adhesion.
Pleurisy	Fibrinous adhesion between the lung and thoracic wall, unassociated with an active pneumonic process.
Pleuropneumonia	Pneumonia associated with inflammation or fibrinous response of the pleura.
Pneumonia	Pneumonic lesions not otherwise considered to meet the definition of "Antero-ventral pneumonia".
Antero-ventral pneumonia ^b	Pneumonic lesions restricted to the antero-ventral aspect of the lung; typically associated with enzootic pneumonia.
Rectal prolapse	Exterior prolapse of the rectal mucosa.
Pyogenic lesion	Presence of external or internal abscess(es).
Septicaemia	Generalised acute (oedematous, hemorrhagic) lymphadenopathy.
Tuberculosis-like lesion	Enlargement of lymph nodes associated with caseous infiltrates.
Wound	Laceration of the skin.

^a Insufficient data available to include in time series analysis.

^b Pneumonia due to infection with *Mycoplasma hyopneumoniae*.

diseases in market weight pigs delivered to commercial abattoirs. The programme, called PigCheck, was developed in 1997 in collaboration with farmers, veterinarians, and a third-party auditing firm that also had responsibility for food safety inspection in those same abattoirs. PigCheck was initially modelled on a similar programme developed in Australia called the Pig Health Monitoring Scheme (Pointon *et al.* 1987), that was later modified for use in the United States of America under the name of PigMON (Pointon *et al.* 1992; Davies *et al.* 1996).

The objective of this project was to complete a retrospective analysis of the existing data from the PigCheck recording system with the specific aims of establishing the prevalence of post-mortem disease lesions, describing any long-term trends in the prevalence of these lesions, and identifying the percentage of the monthly variation in lesion prevalence that could be attributed to individual farms and abattoirs.

Materials and methods

The occurrence of 20 pre-defined carcass and viscera lesions (Table 1) is recorded in real-time on every pig slaughtered in the five largest pig abattoirs in New Zealand. Data from PigCheck are recorded for approximately 94% of commercial pigs in the country (P. Turton, pers. comm.¹).

Data from PigCheck are available to individual farmers and their veterinarians within 1 to 2 days of collection through pre-formatted online reports, but farmers are constrained to viewing only data from their own farms. No formal national reporting from the PigCheck system has previously been carried out,

resulting in a limited perspective on the overall health status of commercial slaughter pigs in New Zealand.

With permission of NZ Pork, a complete download of all PigCheck data collected between January 2000 and December 2010 was obtained. The data for this study were held and manipulated in Microsoft Excel (Microsoft Corporation, Redmond, WA, USA) with further analyses carried out as described below. Slaughter lesion data were collected and reported at the lot level (a lot of pigs is defined as a cohort of pigs delivered from one farm, at one time); no individual pig level or farm location data were available. The data were examined and criteria were developed to ensure only reliable data were retained for further analysis: Lots of more than 1,000 head were removed as these were believed to represent either data entry errors, or multi-day data aggregations with loss of farm-level detail. Lots with zero animals were removed, as were lots when the percentage of affected pigs (for any recorded lesion) was calculated to be greater than 100% as these were assumed to represent data entry errors. Lesion data were reported in a dichotomous manner with the lesion being recorded as either present or absent in each carcass.

A total of 279 unique farms were represented in the dataset. Farms that only supplied pigs intermittently for slaughter or that supplied pigs in very small numbers (one or two pigs) were often not assigned a unique farm identifier in the PigCheck system. These farms were assigned a generic identification number and for the current analysis, their data were treated as if the pigs originated from a single farm, represented as the 280th unique farm in the dataset.

Statistical analyses

All statistical analyses were completed in R version 2.14.1 (R Development Core Team 2011; R Foundation for Statistical Computing, Vienna, Austria).

¹ P Turton,ASUREQuality, Ashburton, New Zealand

Differences in lot sizes between abattoirs were analysed using ANOVA.

Time-series analysis of data for each lesion was conducted after aggregating lesion prevalence for each month of the study. The prevalence of each lesion was regressed on month to determine if a trend was present over the time period. When the slope of the fitted regression line was significantly different than zero ($p < 0.05$), the increasing or decreasing trend was considered to be significant. When a significant trend was identified, the time series was detrended and the time series pattern for the prevalence of each lesion was described by applying an auto-regressive integrated moving average (ARIMA) model to the detrended data. The ARIMA models were fitted using an automated procedure, and then 95% CI for the coefficients of each model term calculated. When the 95% CI for the coefficient of any term contained zero, it was dropped from the model and the ARIMA model fitting procedure was completed manually. The overall fit of each time series model was assessed using Akaike Information Criteria (Diggle 1990); model residuals were examined by plotting them as a time series, assessing the autocorrelation function, and applying the Ljung-Box test for independence (Ljung and Box 1978). To evaluate the presence of any seasonal pattern in the prevalence of each lesion, a frequency analysis of the aggregated monthly data was conducted and the results were reported in the form of a periodogram. Each periodogram was subjectively inspected to identify the occurrence of peaks suggestive of a seasonal pattern.

The prevalence of pigs reported with each lesion was expected to vary between farms and between abattoirs. Over the period of data collection, different farms delivered pigs to different abattoirs creating a nested relationship in the data. To determine the percentage of variance in lesion prevalence that could be attributed to farms relative to that attributed to abattoirs, a hierarchical binomial generalised linear mixed model was created. Lot was weighted by number of pigs in each lot. Two random effect levels were incorporated at the farm (within abattoir) and abattoir levels. Only data from the 279 farms with known unique identity were used in the analysis. Using a GLM with a binomial linking function, the prevalence of each lesion in a lot was regressed on farm within abattoir and weighted by the number of pigs in the lot, using the following equation:

$$\text{For lot } i, \text{ in farm } j \text{ in abattoir } k, \text{ logit}(p_{ijk}) = \text{tv } \alpha + u_{jk} + v_k$$

where u_{jk} and v_k were farm and abattoir level random effects, respectively.

Results

Summary information

After the dataset extracted from PigCheck was cleaned as described above, the final dataset created for analysis contained 124,407 records and described the presence or absence of lesions on 6,220,664 pigs. Data for the analysis were collected from January 2000 through December 2010.

Across the five abattoirs represented in the dataset, lesion information was collected from a mean of 47,126 (95% CI=46,051–48,202) pigs each month. Three of the five abattoirs (abattoirs A, C, and D) provided 83% of the data. The mean lot size was 50.0 (95% CI=49.7–50.3) pigs with significant differences in mean lot sizes noted between some abattoirs (Table 2); however the mean lot size only ranged from 46.5 to 52.3 pigs between abattoirs, suggesting that in practical terms lot size was very similar between abattoirs. Lesions were recorded from a mean of 942 (95% CI=920–965) lots of pigs each month of the study.

Time series analysis

Of the data recorded on the occurrence of the 20 lesions, the occurrence of emaciation, full gut, and pericarditis lesions were sparse or intermittent so they were not evaluated as part of the time series analysis. Three lesions formed a cluster of the most prevalent conditions recorded across the 11-year time series (Table 3); these were antero-ventral pneumonia, pleuropneumonia and milk spots, having mean prevalences of 17.6, 11.4 and 9.2%, respectively. A second cluster of four common lesions was identified, although at a much lower prevalence; these were rectal prolapse, mange, pleurisy and septicaemia, with mean prevalences of 5.8, 3.6, 3.1 and 2.7%, respectively. All other lesions were reported at <2% prevalence.

A significant annual change in prevalence (trend) was identified for every condition except pyogenic lesion, tuberculosis-like lesion, other lesions, and kidney condition (Table 3). However, given the overall low prevalence of lesions it followed that the actual values for these annual trends were very small. The lesions with the most substantial annual decrease in prevalence were milk spots (decreasing 0.54 percentage points per year), pleuropneumonia (decreasing 0.97 percentage points per year), and antero-ventral pneumonia (decreasing 0.73 percentage points per year). By contrast, pleurisy (increasing 0.29 percentage points per year) and mange (increasing 0.23 percentage points per year) were identified as the lesions with the most substantial increases in prevalence. Of the 15 lesions with significant

Table 2. Summary of number of lots (cohorts of pigs from one farm, at one time) of pigs delivered to five abattoirs in New Zealand from January 2000 to December 2010

Abattoir ID	Number (%) of lots	Total pigs delivered	Mean lot size	Number of suppliers
A	40,493 (33)	2,091,855	51.7 ^x	159
B	10,413 (8)	545,168	52.3 ^x	65
C	29,993 (24)	1,394,805	46.5 ^y	28
D	32,314 (26)	1,635,454	50.6 ^z	53
E	11,194 (9)	553,382	49.4 ^z	22
Total	124,407 (100)	6,220,664	50.0	280 ^a

^a Total number of unique suppliers to all abattoirs. Some suppliers delivered pigs to more than one abattoir.

^{xyz} Differences between means in each row with different subscripts are significant (Tukey's HSD, $p < 0.05$).

Table 3. Estimated overall prevalence of 17 lesions reported in a national abattoir-based lesion recording system between January 2000 and December 2010 among 279 pig farms in New Zealand. The mean annual change in prevalence (trend) for each lesion is presented, including the autoregressive, integrated, and moving average (ARIMA) features of a time series analysis.

Lesion name	Months ^a	Prevalence (95% CI)	Trend		
			Annual (%)	p-value	ARIMA
Arthritis	132	1.85 (1.83–1.88)%	+0.06	<0.001	1,0,0
Broken back	132	0.62 (0.6–0.63)%	–0.03	0.001	2,1,0
Bruising	132	1.35 (1.33–1.37)%	–0.04	<0.001	2,2,0
Ileitis	132	0.55 (0.51–0.58)%	+0.10	<0.001	1,1,0
Kidney condition	112 ^b	0.15 (0.15–0.16)%	–0.01	0.050	1,0,0
Mange	132	3.63 (3.58–3.68)%	+0.23	<0.001	2,1,0
Milk spots	132	9.21 (9.09–9.32)%	–0.54	<0.001	0,0,1
Other lesions	132	0.08 (0.07–0.09)%	N/A	0.859	1,1,1
Pleurisy	132	3.13 (3.1–3.16)%	+0.29	<0.001	1,0,0
Pleuropneumonia	132	11.43 (11.33–11.53)%	–0.97	<0.001	1,0,0
Pneumonia	132	0.91 (0.89–0.94)%	–0.12	<0.001	2,1,1
Antero-ventral pneumonia	132	17.57 (17.45–17.7)%	–0.73	<0.001	1,1,1
Rectal prolapse	132	5.77 (5.7–5.84)%	–0.33	<0.001	1,1,1
Pyogenic lesion	132	0.01 (0–0.01)%	N/A	0.106	1,1,0
Septicaemia	132	2.69 (2.65–2.73)%	–0.04	<0.001	2,1,0
Tuberculosis-like lesion	132	0.01 (0.01–0.01)%	<–0.01	0.050	0,0,0
Wound	132	0.95 (0.92–0.98)%	+0.17	<0.001	1,1,0

^a Data available from January 2000 through December 2010.

^b Data available from September 2001 through December 2010.

annual trends in prevalence, 10 were decreasing and five were increasing.

The occurrence of each lesion was evaluated to determine the existence of any seasonal patterns. After detrending the data, conducting a frequency analysis and evaluating the periodogram, no seasonality was identified in 15 of the lesions. However, milk spot and pleurisy were each found to have a distinct seasonal pattern (Figure 1). The prevalence of milk spot tended to peak during late-winter (July and August) particularly during the first two-thirds of the time series. However, as the overall prevalence of milk spot trended down during the period of study, the seasonal pattern became less apparent. The prevalence of pleurisy peaked during early-summer (November and December) although the pattern was difficult to detect visually on the time series plot due to the overall low prevalence of the lesion.

The farm-level prevalence of the 17 lesions that underwent a time series analysis is summarised in Table 4. Across the 11 years for which data were collected, more than half of farms were identified as having at least one pig positive for every condition except for ileitis, pyogenic lesion, and tuberculosis-like lesion. The most prevalent condition between-farms was milk spot, with a mean of 91.8 (95% CI=87.9–94.7)%.

The three ARIMA parameters (autoregressive, integrated, and moving average) associated with the prevalence of each disease lesion are presented in Table 3. This information, when combined with the mean value for a lesion, is sufficient for estimating future values which would allow one to correctly interpret monthly slaughter lesion data collected as part of a future national disease surveillance programme. Without knowledge of the longitudinal pattern of disease occurrence, particularly when seasonal effects are known, it is very difficult to know whether the measured prevalence of a disease lesion is truly 'high', 'low', or 'normal'.

Variance components

By examining the variance associated with farm and abattoir terms in the mixed model, the relative contribution of each to the observed differences in the overall prevalence of each lesion was determined. A summary of the findings is presented in Figure 2.

The lesions for which the abattoir, relative to the farm, explained a greater ($p<0.05$) percentage of variance in prevalence included pyogenic lesion (92 (95%CI=79–100)%), mange (73 (95% CI=69–77)%), and ileitis (62 (95% CI=46–78)%). The lesions for which the farm explained a greater ($p<0.05$) percentage of variance in prevalence included rectal prolapse (98 (95% CI=70–100)%), pneumonia (97 (95% CI=87–100)%), antero-ventral pneumonia (96 (95% CI=87–100)%), arthritis (96 (95% CI=79–100)%), pleuropneumonia (94 (95% CI=88–100)%), and septicaemia (92 (95% CI=81–100)%). A significant amount of the variance in prevalence of broken back, bruising, pleurisy, milk spot and wound lesions was also explained by farm (versus abattoir) but the effect was less than noted for the lesions described above. For tuberculosis-like lesion, kidney condition, and other lesions the 95% CI for farm-attributed and abattoir-attributed variance overlapped, so the effect was considered to be unbiased or not influential.

Discussion

The 279 unique farms analysed in this study represented >90% of the commercial pig production in the country from 2000 to 2010 and can therefore be considered to establish accurate benchmarks for the important production limiting diseases in New Zealand.

Antero-ventral pneumonia (prevalence 17.6%) and pleuropneumonia (11.4%) were the most prevalent lesions in the data set.

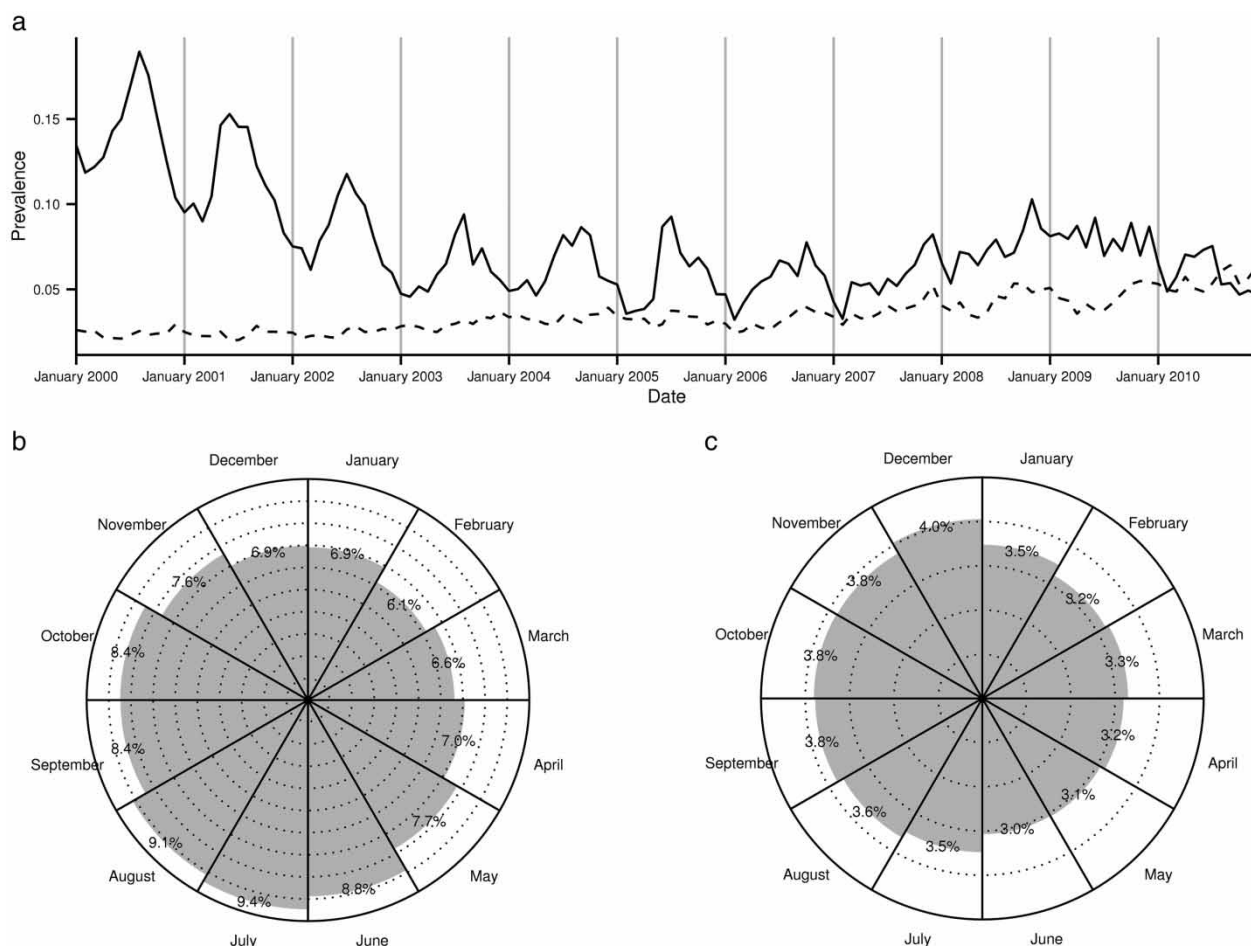


Figure 1. Time series plot (a) of the prevalence of milk spots (solid line) and pleurisy (dashed line) lesions in New Zealand pigs from January 2000 to December 2010. Mean prevalence of milk spots (b) and pleurisy (c) by month is shown in polar plots.

Pleurisy lesions (prevalence 3.1%) were also routinely identified in New Zealand pigs. Aside from their contribution to increased costs of production, lung pathology (particularly pleuropneumonia and pleurisy) creates problems in abattoirs because many of these carcasses require extra trimming and labour input during

Table 4. Between-farm level prevalence of 17 lesions reported in a national abattoir-based lesion recording system between January 2000 and December 2010 among 279 pig farms in New Zealand

Lesion name	Prevalence (95% CI)
Arthritis	86.4 (81.8–90.2)%
Broken back	75.3 (69.8–80.2)%
Bruising	85.7 (81.0–89.6)%
Ileitis	36.2 (30.6–42.1)%
Kidney condition	64.2 (58.2–69.8)%
Mange	84.2 (79.4–88.3)%
Milk spots	91.8 (87.9–94.7)%
Other lesions	57.7 (51.7–63.6)%
Pleurisy	86.7 (82.2–90.5)%
Pleuropneumonia	90.0 (85.8–93.2)%
Pneumonia	71.7 (66.0–76.9)%
Antero-ventral pneumonia	88.8 (83.8–91.7)%
Rectal prolapse	86.7 (82.2–90.5)%
Pyogenic lesion	9.0 (5.9–12.9)%
Septicaemia	84.6 (79.8–88.6)%
Tuberculosis-like lesion	24.4 (19.5–29.8)%
Wound	71.0 (65.3–76.2)%

processing. In the current study, the highest prevalence of pleurisy occurred in the spring. Though the existence of a seasonal effect was unequivocal, the magnitude of the effect only amounted to approximately 1% increase in prevalence during the peak months.

Few studies of abattoir lesions in pigs raised in the southern hemisphere have been published in recent years. A study of abattoir lesions in New Zealand conducted from 1986 to 1990 reported the prevalence of enzootic pneumonia (cranio-ventral pneumonia typically associated with *Mycoplasma hyopneumoniae* infection) to be 45% (Christensen and Cullinane 1990). A later longitudinal study of pig farms in New Zealand was undertaken between winter 1995 and summer 1996 to identify risk factors for respiratory disease. Winter prevalence of enzootic pneumonia was 63.4%, pleuropneumonia 2.7%, and pleurisy 19.1%. Enzootic pneumonia was significantly lower (52.2%) in the summer than in the winter (Stark *et al.* 1998). A study of enzootic pneumonia in abattoir pigs in South Australia was conducted in 1980 and found that 45.1% of lungs had enzootic pneumonia lesions and there was substantial variation in prevalence among the three abattoirs in the study. Prevalence of pleurisy was 14.5% and again there was substantial variation among abattoirs (Pointon and Sloane 1984). In South Australia and Western Australia, formalised slaughter monitoring services commenced in the mid 1980s. Across 157 herds in Western Australia in 1987, approximately 10% of pigs had pleuropneumonia, 20% had pleurisy, and 40% had lesions of enzootic pneumonia (Pointon *et al.* 1987; Mercy and Brennan 1988).

While lesions related to respiratory disease were amongst the most prevalent in the current New Zealand data, it appears that the

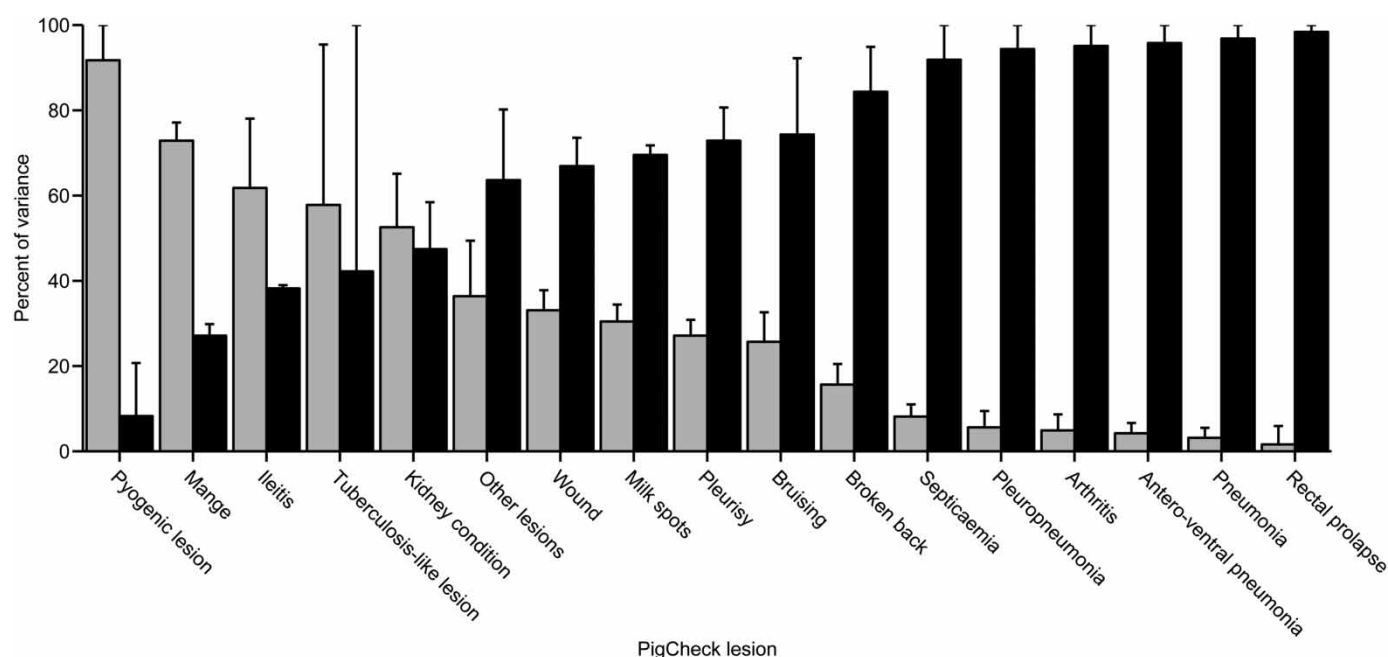


Figure 2. Percentage (with 95% CI) of variance in lesion prevalence attributable to farm (black bars) or abattoir (grey bars) for 17 lesions reported in a national abattoir-based lesion recording system between January 2000 and December 2010. The information was derived from data collected at five abattoirs from 279 New Zealand farms.

overall frequency of their occurrence has reduced from prior surveys and compared favourably to similar data generated from Australia.

Milk spot lesions were identified as the third most prevalent lesion (9.2%) in this study. This term refers to the whitish-coloured foci that occur in the liver stroma and are associated with a healing wound resulting from the migration of *Ascaris suum* larvae. Similar to the situation with lung lesions, few studies of the occurrence of ascariasis lesions in the southern hemisphere are available. However, milk spots were observed on 13% of livers that were examined in a 1990 New Zealand study (Christensen and Cullinane 1990). Milk spots only persist for approximately one-month post-migration of the larvae, making the lesion a sensitive indicator of recent exposure to the parasite (Stewart and Hoyt 2006). Ascariasis in New Zealand pigs occurred at a low level in the present study and appears to be decreasing over time.

Most studies of abattoir lesions focus on lung pathology with comparatively fewer studies available that include information on the other lesions that were described in the PigCheck data. The prevalence of ileitis (7%) and mange (13%) in slaughter pigs was previously reported in New Zealand (Christensen and Cullinane 1990), both of which were higher than the estimates of 0.6 and 3.6%, respectively, reported here.

Data from this study provide insight into the relative contribution of the farm versus the abattoir in the variation of lesion prevalence. Variation in lesions that were more difficult for inspectors to identify (or that simply had a greater degree of normal variation), such as mange and ileitis, were strongly influenced by the abattoir at which the pigs were processed. Easy to identify lesions such as rectal prolapse and various lung pathologies were more strongly influenced by farm factors. These findings will be important in any future revisions of the lesion definitions, for inspector training, and in order for farmers and veterinarians to correctly interpret PigCheck results. It should be noted that definitions and descriptive guides that document the criteria being evaluated for each lesion being assessed in PigCheck were not

readily available to the authors. It would be appropriate and timely for this information to be disseminated routinely to Pig-Check customers; further value from the system could be derived by formally reviewing the definitions every 2 to 3 years.

Abattoir monitoring programmes are not without their deficiencies and various authors have described limitations in some of the historical and current programmes. Concerns that have been identified include the fact that the sampling frame is biased given most, if not all, animals delivered to an abattoir are clinically healthy, that few of the lesions being recorded are pathognomonic for a particular disease thus limiting the specificity of the surveillance activity, and that the high speed of modern abattoirs makes complete and accurate evaluation of tissues very difficult (Christiansen and Hellstrom 1979). Other authors have suggested limitations related to the sustainability of abattoir surveillance programmes, as the willingness of a farmer to pay for the service may decline as the overall health of his stock improves, and that visits by veterinarians prompted by abattoir findings may not be timely and in fact may occur after the farmer has already instituted an intervention (Christensen *et al.* 1994). In a study that reviewed the effectiveness of four concurrently operating surveillance systems in the English pig industry, two of which utilised samples or observations at abattoirs, the investigators observed that there existed a general lack of integration of the monitoring and surveillance systems both at the strategic and the operational levels, that there was a lack of routine in-depth analysis of the data, and that there was an unknown level of bias in the data being collected (Stark and Nevel 2009). All of these findings would be important considerations in making future improvements to the programmes.

However, despite their shortcomings there is evidence that abattoir monitoring programmes can be a useful part of an early warning system for animal diseases. Canadian researchers completed a retrospective analysis of abattoir data to determine if there were clues that could have led to early detection of a large outbreak of disease caused by porcine circovirus type 2 in

Ontario (Thomas-Bachli *et al.* 2012). The investigators determined that an increasing occurrence of kidney lesions was spatially and temporally related to the outbreak, although pneumonia lesions appeared not to be related.

Little research has been conducted to assess the utility of abattoir surveillance to the farmer or veterinarian. In a 1985 survey of Canadian farmers and veterinarians, 89.4% said they intended to use the disease information produced by a computerised reporting system of abattoir lesions (Shadbolt *et al.* 1987). These same farmers (50.3%) and veterinarians (57.1%) reported that they would like the information on a monthly basis and that the information be as detailed as possible, including information about disease severity not just disease frequency. However, 56.3% of farmers in the study also disclosed that they actually never discussed the abattoir surveillance results with their veterinarian. In contrast, 60.3% of farmers believed that summary information representing all farms in the country or region should be shared with industry veterinarians.

In summary, the overall prevalence of most lesions recorded in the PigCheck system for the period 2000 through 2010 was low relative to published data from other countries. Common lung pathologies contributing to lesions such as pneumonia and pleurisy were primarily a function of farm management and were not likely due to variability in lesion recording at different abattoirs. Based on the low frequency of lesions identified in pigs at commercial abattoirs, the health status of the New Zealand pig industry was very good. Pneumonia, pleurisy, and ascariasis were some of the most prevalent conditions that should be focussed on through development of herd health management plans.

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